

Algorithms Problem Solving Level 1

Commission Percentage

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and tech-oriented.

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Commission Percentage

Definition

- ◆ This program reads the user's total sales and calculates the commission based on the sales amount.
- ◆ Different sales ranges have different commission percentages.
- ◆ The program first determines the commission rate, then multiplies it by the total sales.

Explanation

- ◆ ``readTotalSale()`` reads the total sales amount and returns it as a ``double``.
- ◆ ``getCommission()`` determines the commission rate according to the total sales.
- ◆ The function returns decimal values such as ``0.01`` for 1% and ``0.05`` for 5%.
- ◆ ``calculateCommission()`` multiplies the total sales by the commission rate.
- ◆ For example:
$$\texttt{`100000 \times 0.03 = 3000`}$$
- ◆ ``printResult()`` prints the calculated commission.
- ◆ ``calculateCommission(readTotalSale())`` first reads the sales amount, calculates the commission, then passes the result to ``printResult()``.

Steps in main()

1. Read the total sales.
2. Check which sales range the value belongs to.
3. Return the corresponding commission rate.
4. Multiply total sales by the commission rate.
5. Return the commission amount.
6. Print the commission.

💡 Example

◆ If the user enters:

Total Sales: 600000

◆ The sales amount is between 500000 and 1000000.

◆ The commission rate is 2%.

◆ Calculation:

$$600000 \times 0.02 = 12000$$

➡ Result

Commission = 12000

Note

◆ The commission rates are represented as decimal values:

1% → `0.01`

2% → `0.02`

3% → `0.03`

5% → `0.05`

0% → `0.00`

◆ The conditions are checked from the highest sales range to the lowest.

⊘ Warning

- ◆ The comment says ``>1000000``, but the code uses ``>= 1000000``.
- ◆ Therefore, exactly 1,000,000 receives a 1% commission in the current code.
- ◆ Similarly, the code treats exactly 500,000 as 2%, exactly 100,000 as 3%, and exactly 50,000 as 5%.



Comparison

Higher Sales Range

- ✓ `>= 1000000` → 1%
- ✓ `>= 500000` → 2%
- ✓ `>= 100000` → 3%
- ✓ `>= 50000` → 5%

Lower Sales Range

- ✓ `< 50000` → 0%
- ✓ No commission is calculated.
- ✓ `calculateCommission()` returns 0.

✦✦ Summary

- ✓ The program calculates commission based on total sales.
- ✓ `getCommission()` returns the appropriate commission rate.
- ✓ Decimal values such as `0.02` represent percentages such as 2%.
- ✓ `calculateCommission()` multiplies sales by the selected rate.
- ✓ The current code uses inclusive boundaries with `>=`.
- ✓ The commission is calculated using one rate on the entire sales amount.

Thank you 😊